

Ansh Tripathi

B.Tech CSE (AI/ML) · LPU Phagwara · 4th Semester

Passionate Computer Science student specializing in AI/ML with strong foundations in DSA, systems, and real-world problem solving through scalable software projects.

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Education

Lovely Professional University

Punjab, India

B.Tech – Computer Science & Engineering (AI/ML Specialization) | CGPA: 7.4

July 2024 – May 2028

- Recipient of **LPUNEST Category 1 merit scholarship** – 30% tuition fee waiver.
- Relevant Coursework: Data Structures & Algorithms, OOPS, DBMS, Python, Discrete Mathematics, AI/ML.
- Actively practicing DSA in C++ and Java on coding platforms to strengthen problem-solving skills.

Saraswati Vidya Mandir

Basti, Uttar Pradesh

Intermediate – PCM + Computer Science | 72%

March 2024

Little Flower School

Siddharth Nagar, Uttar Pradesh

Matriculation | 91%

March 2022

Technical Skills

Languages: C, C++, Java, Python, JavaScript, HTML/CSS

Databases: PostgreSQL, SQL

ML & Data Science: NumPy, Pandas, Matplotlib, Scikit-learn (basic), Jupyter Notebook

Frameworks & Tools: React, TailwindCSS, Flask, FastAPI, Git, GitHub, VS Code, Google Colab

CS Concepts: Data Structures & Algorithms, OOPS, OS Scheduling, Real-Time Systems, RESTful APIs

Projects

SmartGrid-GOA-LoadBalancing

March 2026

Python · Grasshopper Optimization Algorithm · Machine Learning · MIT License

- github.com/ansh10tripathi/SmartGrid-GOA-LoadBalancing
- Designed an ML-based smart grid load balancing system using Grasshopper Optimization Algorithm (GOA) to optimize energy distribution.
- Simulated dynamic allocation across multiple grid nodes, reducing power imbalance and improving system stability.
- Integrated machine learning with bio-inspired optimization to solve real-world energy management problems.

Chemical-Solubility-ML-Model

February 2026

Python · Scikit-learn · Pandas · Matplotlib · Jupyter Notebook

- github.com/ansh10tripathi/Chemical-Solubility-ML-Model
- Built a regression pipeline to predict chemical solubility (LogS) using Linear Regression and Random Forest models.
- Evaluated model performance using R^2 , RMSE, and MAE, with feature importance analysis.
- Designed a complete ML workflow including EDA, preprocessing, training, and visualization in Jupyter Notebook.

Priority-Inversion-Handling-System

March 2026

Python · Real-Time OS Simulation · Scheduling Algorithms

- github.com/ansh10tripathi/priority-inversion-handling-system
- Simulated priority inversion in real-time systems and built PIP and PCP protocols to resolve scheduling conflicts.
- Engineered concurrent task execution with mutex locks, ensuring correct priority handling.
- Applied OS concepts like semaphores, synchronization, and scheduling algorithms.

BidWise-RTB-Engine

February 2026

TypeScript · AI Optimization · Real-Time Bidding · Auction Simulation

- github.com/ansh10tripathi/BidWise-RTB-Engine
- Engineered an AI-based real-time bidding engine simulating DSP behavior with second-price auction logic.
- Built adaptive bidding strategies based on budget, conversion probability, and user data.

- Engineered ROI optimization and dynamic budget allocation in ad-tech scenarios.

Productivity-Hub

February 2026

React · TailwindCSS · JavaScript · MIT License

- github.com/ansh10tripathi/productivity-hub
- Designed a responsive task management web app with drag-and-drop, dark mode, and filtering features.
- Implemented component-based architecture using React and TailwindCSS with persistent state storage.

Hackathons & Competitions

- **Adobe India Hackathon** – Cleared Round 1 Online MCQ Assessment (Certificate awarded).
- **Trinetra Hackathon** – Participated (Certificate of Participation).
- **Just In Case '25** – The Global Case Competition (Certificate of Participation).
- **All AI Tools Tech Challenge** – Certificate of Participation; completed AI Tools Workshop.

Certifications

- Linux Fundamentals – Skillera | Sept 2024 – Nov 2024
- AI Tools Workshop – Completed (2025)

Leadership & Activities

- Active member of **Excel to Accelerate** club, contributing to technical and collaborative initiatives.
- Part of team **Apex**, working on group-based technical projects and problem-solving tasks.

Achievements

- **LPUNEST Category 1** – Top merit scholarship tier in national-level engineering entrance examination (30% fee waiver).
- **10+ active public GitHub repositories** with MIT licensing and structured READMEs maintained since July 2024.
- Inter-school athletics – **Shot Put, Javelin, Volleyball, Football** – competed at district level.